

Christie in the M1931 tank he designed for the US Army



Great designers J. Walter Christie

J. Walter Christie is often recorded as being a maverick inventor—irascible, argumentative, and hard to deal with—and it may be that his personality got in the way of his tank designs seeing series production. However, some of his inventions had a great influence on tank development.

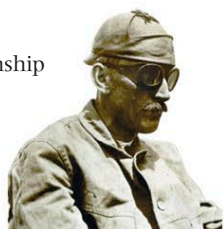
CHRISTIE, AN AMERICAN, worked as a consulting engineer for a number of steamship lines before turning his attention to car racing. He designed and drove a front-wheel-drive car in the 1907 French Grand Prix, and later the same year was involved in a serious car accident at a racetrack in Pittsburgh while attempting to set a track record.

A car he designed—the Christie Racer—later became the first car to lap the Indianapolis speedway at over 100mph (161 km/h).

Christie also designed taxicabs and fire engines. During World War I, he designed a gun carriage for the US Ordnance Board, but refused to listen to his client's specific requirements.

Experimental design

Christie's experimental T3E2 tank could move fast, but only had room for a crew of two. The US Army deemed it inadequate as an infantry support weapon.



J. Walter Christie
(1865-1944)

His stubbornness in arguing his case and his rudeness before authority became a pattern that did not endear him to the military authorities. However, he did have some success with an amphibious light tank that the US Marine Corps thought had potential, even though it initially had trouble reaching shore

during a test. He developed his interest in tanks, and after years of experimentation and large financial investment he displayed a radical new tank chassis to the US military in October 1928. Christie wanted to call his tank Model 1940, because he considered it a dozen years ahead of its time, but it was designated Model 1928.

The new vehicle had large road wheels, which it could run on with the tracks removed. Uniquely, each wheel had its own independent suspension—



Passing the test

A T3E2 tank with Christie suspension crosses an obstacle course in 1936. Each wheel had its own suspension, enabling the tank to cross difficult ground with ease.

a “helicoil” spring mounted inside the hull—which gave the tank an extraordinary agility when crossing rough ground, each wheel bumping up and down as it passed over an obstacle. This enabled the tank to travel much faster than conventional tanks, which used the relatively cumbersome “leaf spring” suspension system. To keep weight down and speed up, the tank had thin armor, which was also sloped at the front to deflect projectiles. Christie envisaged his tank being used to penetrate enemy positions and travel at speed far into enemy territory. It weighed only 9 tons (8 tonnes) and had a Liberty engine that gave it a top speed of 42 mph (68 km/h) on tracks—and an extraordinary 70 mph (112 km/h) on its wheels.

The US Army Infantry Tank Board was unimpressed by the tank's thin armor—they saw tanks as infantry support weapons—so they passed Christie onto the Cavalry, who were more interested in armored cars at the time. To add to Christie's frustration, the US military refused to pay the development costs he had incurred.

Christie became more argumentative and embittered, and decided to sell his designs to the highest bidder. This led him to having dealings with a number of foreign countries:

